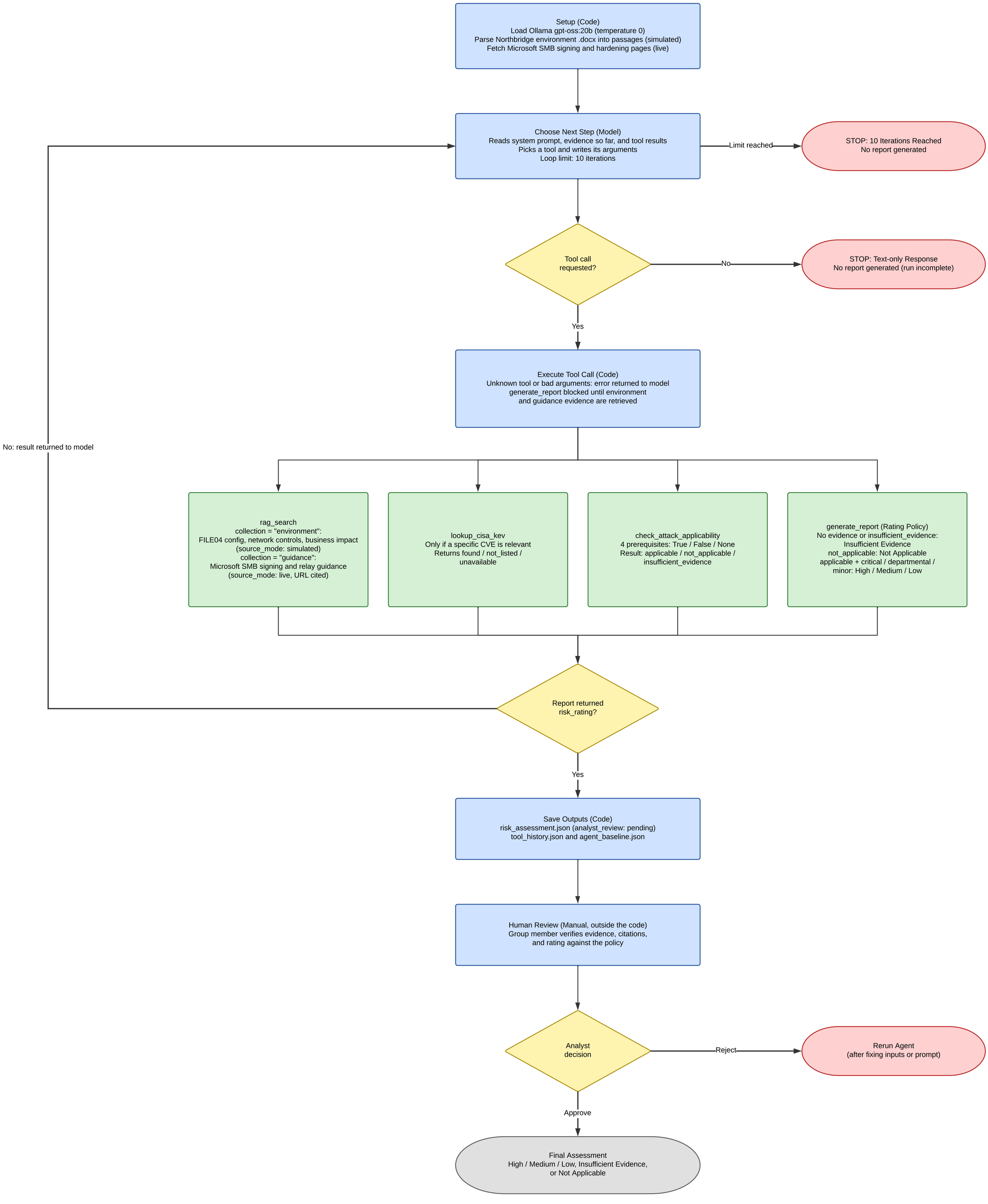
Setup (Code)
Load Ollama gpt-oss:20b (temperature 0)
Parse Northbridge environment .docx into passages (simulated)
Fetch Microsoft SMB signing and hardening pages (live)
Choose Next Step (Model)
Reads system prompt, evidence so far, and tool results
Picks a tool and writes its arguments
Loop limit: 10 iterations
Limit reached
STOP: 10 Iterations Reached
No report generated
Tool call requested?
No
STOP: Text-only Response
No report generated (run incomplete)
Yes
Execute Tool Call (Code)
Unknown tool or bad arguments: error returned to model
generate_report blocked until environment
and guidance evidence are retrieved
No: result returned to model
rag_search
collection = "environment":
FILE04 config, network controls, business impact
(source_mode: simulated)
collection = "guidance":
Microsoft SMB signing and relay guidance
(source_mode: live, URL cited)
lookup_cisa_kev
Only if a specific CVE is relevant
Returns found / not_listed /
unavailable
check_attack_applicability
4 prerequisites: True / False / None
Result: applicable / not_applicable /
insufficient_evidence
generate_report (Rating Policy)
No evidence or insufficient_evidence:
Insufficient Evidence
not_applicable: Not Applicable
applicable + critical / departmental /
minor: High / Medium / Low
Report returned
risk_rating?
Yes
Save Outputs (Code)
risk_assessment.json (analyst_review: pending)
tool_history.json and agent_baseline.json
Human Review (Manual, outside the code)
Group member verifies evidence, citations,
and rating against the policy
Analyst
decision
Reject
Rerun Agent
(after fixing inputs or prompt)
Approve
Final Assessment
High / Medium / Low, Insufficient Evidence,
or Not Applicable